6 Find the particular solution of the differential equation

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 12\frac{\mathrm{d}x}{\mathrm{d}t} + 36x = 37\sin t,$$

given that, when $t = 0$, $x = \frac{\mathrm{d}x}{\mathrm{d}t} = 0$. [11]